Spectroscopy of Gas Lamps using a Spectrometer and Prism

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Abstract

In this experiment, we measured the wavelengths of light from various gas-discharge lamps through two methods: one with a diffraction grating and the other with a prism. Using the diffraction grating, Lab B and Lab C measured the sodium, hydrogen, and deuterium lamps with the diffraction grating, while Lab D measured mercury, hydrogen, and deuterium. The prism analysis focused on the data provided by Lab D, which measured mercury, sodium, and deuterium. The calculations for the diffraction grating utilized the data collected for hydrogen and deuterium to produce a Rydberg constant comparable to the expected value. The Rydberg constant for hydrogen was $107912.014 \pm 5374.38\ cm^{-1}$ for Lab B, $174747.97 \pm 16968.71\ cm^{-1}$ for Lab C, and $143185.71 \pm 22875.62\ cm^{-1}$ for Lab D. In Procedure B, the index of refraction can be calculated from the angles of minimum deviation. The range of values for the index of refraction n is 1.66-1.67, which is approximately the determined n value for the medium of glass (1.52-1.66) [13].

1 Introduction

The objective of this lab is to measure the wavelengths of light from different gas-discharge lamps through two methods: one with a diffraction grating and the other with a prism. Concerning the diffraction grating, the spectrum of sodium, mercury, hydrogen, and deuterium will be recorded. Meanwhile, the prism procedure will measure the angle of minimum deviation from gas-discharge lamps. The data from the diffraction grating, specifically hydrogen and deuterium, will be utilized to produce a Rydberg constant comparable to the expected value. In procedure B, an index of refraction can be calculated from the minimum angles of deviation recorded and calibration from the obtained data.

Although investigations of light may have occurred earlier, the first documented explanation examining the behavior of light was conducted by Isaac Newton. His publication "Opticks" includes a section on "the phenomenon of colors", which conveys the findings from what is now acknowledged as Newton's famous prism experiment [4]. The experiment inspired scientists such as William Herschel, William Wollaston, and Thomas Young to reproduce the diffraction aspect but, with their spin on it. In the early 1800s, Herschel discovered the infrared spectrum, Wollaston investigated dark lines in the solar spectrum, and Young devised his double-split experiment [7]. Subsequently in 1814, Joseph van Fraunhofer observed the dark lines in the solar spectrum similar to Wollaston; however, he could not provide an explanation for this strange phenomenon [1]. The 700 lines observed were categorized and named "The Fraunhofer lines".

The prism method continued as the preferred instrument of use, specifically in astronomy for the classification of stars. Despite this tool's universal application in spectroscopy, it had limitations and often underwent alterations to accommodate advancing topics or experiments in the field. One of the prism's characteristics that was improved upon was the resolving power; the main incentive was to obtain better accuracy and precision when recording data [7]. It was not until about the last century that a spectroscope was invented. Since then, there have been more significant discoveries, such as Bohr's theory of the hydrogen model in

1913. Semiconductors in the 1950s revolutionized spectroscopy and eventually sparked the creation of a grating through interference spectroscopy [7]. This development was favored for its high resolution and precision in observing the structure of spectral lines. In modern-day spectroscopy, much of it is dependent on the evolution of technologies over time, such as the spectroscope and diffraction grating, to contribute to laser optics, quantum spectroscopy, and other current topics.

2 Theory

Spectroscopy is the study of the behavior of light and how different elements interact with light. The optimal instrument to observe this relationship is called a spectroscope or spectrometer, which operates by separating a stream of light emitted by a material into visible colors/lines. Each element has a unique combination of spectral lines when exposed to heat or electricity known as the emission spectrum. It is utilized as an identifier of chemical composition, similar to how crimes can be ascertained through the distinctiveness of human fingerprints.

A spectroscope can identify an element, operating by in-taking light reflected from a material through a thin slot in a metal plate. The beam of light will deflect off of a special grating, revealing the corresponding emission spectra [6]. Since each spectrum will be characteristic of each element, this means no element can share an identical spectrum. This is due to the difference in absorption or emission rates of materials, caused by the various combinations of photons and electrons possible. To visualize, electrons can jump energy levels during absorption and reduce levels during emission [9]. Therefore, the shortest wavelength is attributed to the highest energy, and the longest wavelength for the shortest energy.

To calculate the wavelength, the equation is

$$\lambda = \frac{d \sin \theta}{m} \tag{1}$$

where d is the distance between lines, m is the order, 'θ' is the angle from the central maxima, and 'λ' is the wavelength. In the case of calculating the distance between the spectral lines, Equation 1 can be manipulated so the 'a' variable is isolated, such as

$$d = \lambda \times \frac{m}{\sin\theta} \tag{2}$$

To calculate the Rydberg constant, the Rydberg formula is necessary. The Rydberg equation is capable of predicting hydrogen spectral lines and appears as

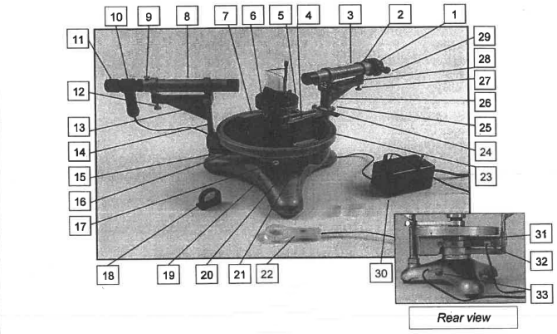
$$\frac{1}{\lambda} = R_H \times \left(\frac{1}{4} - \frac{1}{m^2}\right) \tag{3}$$

where 'λ' is the wavelengths, 'R_H' is the Rydberg constant, and m is the Balmer series order [5]. Lastly, when calculating the index of refraction, or the quantity of how light travels through a medium, the equation

$$n = \frac{\sin\left(\frac{\alpha}{2} + \frac{\delta_{\min}}{2}\right)}{\sin\left(\frac{\alpha}{2}\right)} \tag{4}$$

is ideal, where 'n' is the index of refraction, 'α' is the vertex angle, and '$\delta_{\min}$' is the minimum angle of deviation [3]. This equation is derived from Snell's law and pertains to the prism.

3 Procedures



(a) Spectrometer

PARTS KEY (refer to *Figure 1*)

1. Slit Unit
2. Slit Unit Clamp Screw
3. Collimator Unit
4. Vernier Locking Arm
5. Stage
6. Stage Leveling Screws (3)
7. Vernier (double-ended)
8. Telescope Unit
9. Eyepiece Clamping Screw
10. Abbé Self-Collimating Eyepiece
11. Diopter Compensation Collar
12. Telescope Leveling Screw
13. Telescope Horizontal Alignment Screw
14. Telescope Support Pillar
15. Telescope Fine Adjustment Knob
16. Telescope Locking Arm
17. Telescope Locking Screw
18. Alignment Mirror for Self-Collimating Eyepiece
19. Base
20. Rotating Telescope Mount with Counterweight
21. Divided Circle
22. Illuminated Vernier Magnifier
23. Stage Clamping Screw
24. Vernier Adjustment Knob
25. Vernier Clamping Screw
26. Collimator Support Pillar
27. Collimator Leveling Screw
28. Collimator Horizontal Alignment Screw (behind support pillar)
29. Slit Width Adjustment Screw
30. Transformer
31. Divided Circle Clamping Screw
32. Connector for Autocollimator Illumination
33. Connector for Transformer

(b) Apparatus key

Figure 1: Apparatus [12]

The apparatus prioritized for the experiment was the spectrometer, as seen in Figure 1. The corresponding key characterizes the various screws and units accessible on the instrument. Alongside the apparatus, other essential components of the laboratory set-up are the accessories, the diffraction grating, and the prism. Both of the objects require the spectrometer to be calibrated to accommodate the shape and size of the particular accessory. To calibrate the spectrometer, the light source must be on. The lamp was allotted a minimum of ten minutes to warm up; this step optimizes the strength of light passing through the eyepieces during calibration or experimental observation. After the lamp illuminated a sufficient amount of light, the telescope and collimator units were aligned. This was arranged so the light source, located at the collimator's rear where the slit unit resides, could be visible through the telescope eyepiece. Then, the observing eye can peer through the eyepiece to view a slit of some size. The telescope leveling screw, on the underside of the telescope, and the telescope horizontal alignment screw, on the side of the telescope entry, were utilized to orientate the eyepiece cross-hairs with the slit. The collimator leveling screw and collimator horizontal alignment screw parallel the telescope alignment screws and were used for the purpose of moving the collimator. After the slit was positioned appropriately, an accessory was placed on the stage. The adjustment of the leveling was attained by turning each of the three Stage

Leveling Screws. Once the accessory was diffracting the source light to the direction of the telescope entryway, a spectrum was visible through the eyepiece.

One of the procedures was to measure the gas-discharge lamps with the diffraction grating. The instrument was calibrated to the spectrum of each light source so the n=0 order aligned with 0 degrees. When measuring the spectrum, the telescope would move around the stage so each spectral line could be precisely measured. For each light source, approximately three orders were visible, with the first being the brightest with the most lines and the third being the dimmest with the least amount of lines visible. The preferred method for measurement involved the degree and the vernier scale, which provided more precise readings in arc minutes. In the field of view, a black box was used for reference when collecting three different values: the best, maximum, and minimum angle. The best measurement was taken in the center of the eyepiece, while the maximum and minimum values were recorded at the edges of the black box. The maximum and minimum angles serve as a range of error for the best angle value. In the second procedure, the prism was utilized for data collection. The prism was placed on the stage so the observable spectrum was observable on one side of the 0° center. By rotating the stage with the prism, the spectral lines will shift in one direction until a specific limit is reached and the direction is reversed. The point at which the lines shift back is called the angle of minimum deviation. Using Equation 4, the index of refraction for the prism can be calculated with the measured angle of minimum deviation values.

4 Analysis and Results

4.1 Analysis

The data for the diffraction grating analysis was provided by Lab B, Lab C, and Lab D. All of the data was measured in degrees for each angle and additionally, arcminutes for precision. The measurements in arcminutes were converted to degrees and added to their

bigger, corresponding angle value. The angles in degrees were converted to radians to be compatible with Excel's `sine` operation. Lab B, C, and D used one element to calibrate the data of the two other elements. Lab B and Lab C depended on sodium to calibrate the data from the hydrogen and deuterium gas-discharge lamps, while Lab D used mercury. The known wavelengths of sodium and mercury were inserted into Equation 1 to find the d variable, or the distance between the lines on the grating. This was only applicable to the unknown grating data, since the known grating had a labeled value of 13400 lines per inch. Therefore, this method of calibration was only performed on the unknown grating data. The wavelengths of hydrogen and deuterium from the unknown grating data sets were found by using the calculated d from sodium or mercury. Equation 1 produced the experimental values of wavelength using the hydrogen or deuterium data set and their respective d. The known grating used the d value of 13400 lines per inch (1895.52 lines per nanometer) to calculate the experimental values of each spectral line.

The prism analysis relied on the data set from Lab D. The sodium data set was calibrated with the mercury and deuterium known wavelengths. The known wavelengths of mercury and deuterium were plotted alongside the corresponding angles of minimum deviation to create a graph that can be fitted to a polynomial trendline. The equation of the trendline is a quadratic equation. A typical quadratic equation appears as

$$y = ax^2 + bx + c \tag{5}$$

except in this case, the equation was represented as

$$\delta_{min} = a\lambda^2 + b\lambda + c \tag{6}$$

where δ_{min} is the angle of standard deviation and λ is the wavelength. Equation 6 was solved using the quadratic equation, such as

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4a(c - \delta_{min})}}{2a} \tag{7}$$

where λ is the experimental value for the spectral line and δ_{min} is the measured angle of minimum deviation. The measured δ_{min} values of sodium were plugged in the quadratic equation derived from the trendline and solved for the wavelength λ in the form of Equation 6. In addition to calculating wavelengths, the index of refraction was found for each spectral line. Equation 4 generated a value for an index of refraction n in association with each angle of minimum deviation δ_{min}. The vertex angle α is consistent ($60°$), since the same prism is used throughout the data collection process.

Alongside the index of refraction, the Rydberg constant calculations are another quantitative result of the experiment. The Rydberg constant can be calculated from Equation 8.

$$\frac{1}{\lambda} = R_H \times \left(\frac{1}{4} - \frac{1}{m^2}\right) \tag{8}$$

Equation 8 can be algebraically manipulated so R_H is the isolated variable. This would appear as

$$R_H = \frac{1}{\lambda\left(\frac{1}{4} - \frac{1}{m^2}\right)} \tag{9}$$

where R_H is the Rydberg constant, λ is the measured wavelength, and m is the order.

Error is considered in three forms: in degrees, standard deviation, and percent difference. The error in degrees is the range of experimental error for the measured values. Lab B uses a measured maximum and minimum value for error, while Lab C and D yielded a standard error for all values. The standard deviation is measured in nanometers and characterized as the error for the average wavelength of each color. Similarly, the error for the averaged wavelengths is compared to their expected values to produce an error of difference in percentage.

4.2 Results

The measured values of hydrogen from Lab B, C, and D are compiled into Table 1.

Table 1: Hydrogen Wavelengths from Diffraction Grating

–	Red (nm)	Teal (nm)	Blue (nm)	Violet (nm)	Error (°)
Expected	656.28	486.13	434.05	410.17	–
Lab B	660.6	486.3	451.32	430.53	-0.34/+0.37
Lab C	568.96	414.85	376.16	347.02	0.1
Lab D	588.12	439.17	–	–	0.083

This data was recorded with the unknown grating, calibrated with sodium data for Lab B and Lab C and mercury data for Lab D. The error of the measured data is in units of degrees, where Lab B has an error of minimum value of 0.34 and an error for the maximum value of 0.37, Lab C has an error of 0.1, and Lab D has a value of 0.083. The standard deviation values of the mean wavelength for each spectral line were organized into Table 2.

Table 2: Hydrogen Wavelengths Error (STD)

–	Red	Teal	Blue	Violet
Expected (nm)	656.28	486.13	434.05	410.17
Lab B (nm)	18.4	24.5	41.7	25.26
Lab C (nm)	39.54	38.06	37.72	37.1
Lab D (nm)	3.65	10.4	–	–

Four spectral lines of colors red (656.28 nm), teal (486.13 nm), blue (434.05 nm), and violet (410.17 nm) were visible for Lab B and Lab C. Only the red and teal wavelengths were visible for Lab D. Moreover, the percent difference of the hydrogen spectrum is displayed in Table 3. The percent difference represents the accuracy of the experimental value with the expected wavelength.

Table 3: Hydrogen Wavelengths Error (Percent Diff.)

–	Red	Teal	Blue	Violet
Expected (nm)	656.28	486.13	434.05	410.17
Lab B (%)	4.85	3.9	0.33	0.66
Lab C (%)	14.25	15.82	14.29	16.7
Lab D (%)	10.15	10.96	–	–

Figure 2 is a visual comparison of the expected values and the experimental values of the hydrogen spectrum for Lab B, Lab C, and Lab D. The error bars are derived from the "Error (degrees" column from Table 1. The error values were converted from degrees to nanometers. Lab C and Lab D produced error bars that fail to be visible due to the scaling of the axes on Figure 2.

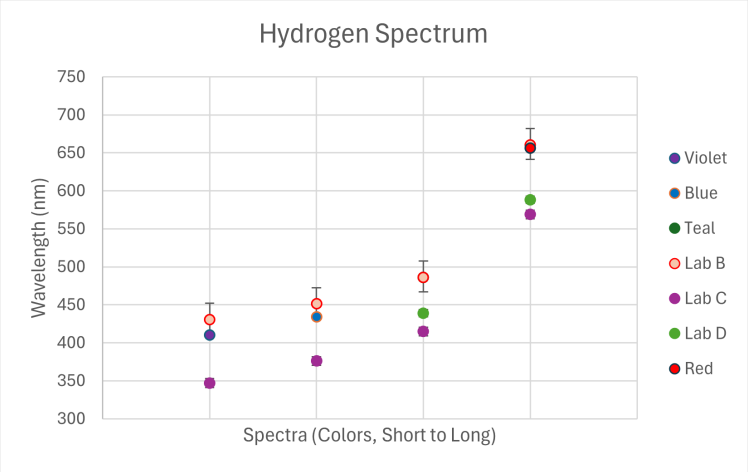


Figure 2: Hydrogen spectrum from diffraction grating.

Table 4 displays the calculated wavelength values of deuterium for Lab B, Lab C, and Lab D.

Table 4: Deuterium Wavelengths from Diffraction Grating

–	Red (nm)	Teal (nm)	Violet (nm)	Error (°)
Expected	656.29	486.14	434.05	–
Lab B	650.15	476.54	432.03	-0.31/+0.47
Lab C	584.36	414.66	350.33	0.1
Lab D	599.94	441.26	395.3	0.083

The deuterium data was calibrated using sodium data for Lab B and Lab C and mercury data for Lab D. Three spectral lines of red (656.29 nm), teal (486.14), and violet (434.05 nm) were observed in all data sets. Lab B produced an error for a minimum value of 0.31 degrees and a maximum value of 0.47 degrees, Lab C has an error of 0.1 degrees, and Lab D has an error of 0.083 degrees. Table 5 are the errors for the average value for each wavelength and

Table 6 are the percent differences between the expected and experimental values for each lab.

Table 5: Deuterium Wavelengths Error (STD)

–	Red	Teal	Violet
Expected	656.29	486.14	434.05
Lab B (nm)	3.54	13.24	1.9
Lab C (nm)	38.44	36.83	34.87
Lab D (nm)	3.82	0.039	1.8

Table 6: Deuterium Wavelengths Error (Percent Difference)

–	Red	Teal	Violet
Expected (nm)	656.29	486.14	434.05
Lab B (%)	0.94	2.00	0.47
Lab C (%)	11.6	15.87	21.35
Lab D (%)	8.97	9.70	8.97

The comparison of the expected values of wavelength and the experimental values for deuterium is represented by Figure 3.

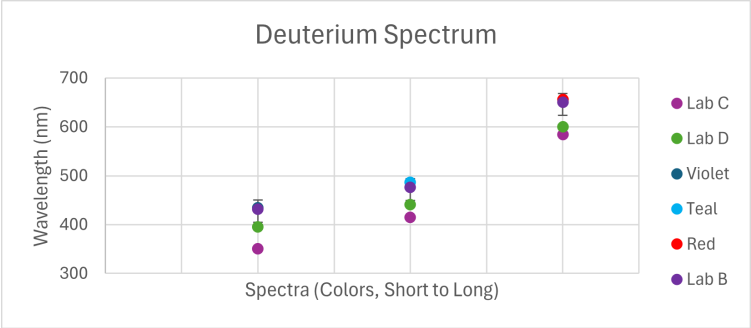


Figure 3: Deuterium spectrum from the diffraction grating.

The calculated wavelengths of the sodium spectrum through a known grating are seen in Table 7.

Table 7 depicts the calculated wavelength and errors for the observed wavelengths yellow (one line of the doublet, 589.6 nm) and green (568.27 nm). No calibration was necessary since the distance between lines is known. The errors measured are the experimental errors

Table 7: Sodium Spectrum for Diffraction Grating, Known

–	Yellow	Green
Expected (nm)	589.6	568.27
Lab B (nm)	607.72	585.75
Error (°)	-11.79/+26.59	-14.87/+4.69
Percent Diff. (%)	13.29	0
STD Error (nm)	3.027	3.029

in degrees, the percent difference, and the deviation from the average wavelength for each spectral line.

The data of Table 7 is represented by Figure 4. Figure 4 is a comparison of the expected wavelength values and the experimental values for Lab B, Lab C, and Lab D. The error bars are determined by the error values (minimum and maximum). The error values were converted from degrees to nanometers.

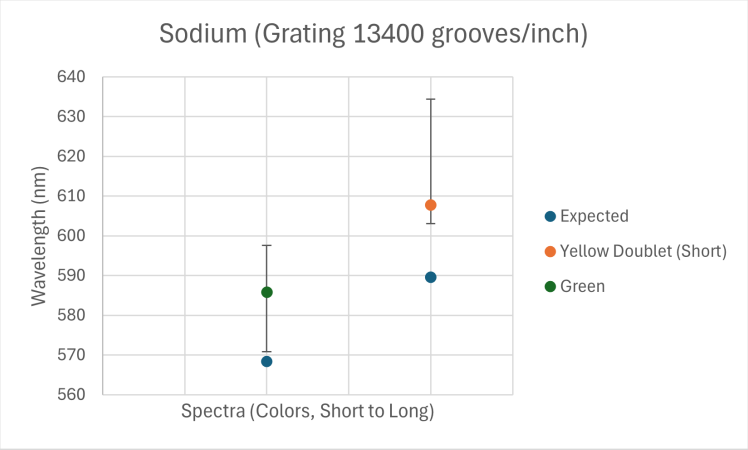


Figure 4: Sodium wavelengths from a known grating.

The wavelengths of sodium that were measured with a prism are compiled into Table 8. These measurements are from Lab C. Lab C provided data from a mercury and deuterium gas-discharge lamp, which was used to calibrate the sodium data.

The process of calibrating the sodium data is illustrated in Figure 5.

Figure 5 is a θ, or angle, vs. wavelength plot. Specifically, the angle θ is the angle of minimum deviation δ_{min}. The known wavelengths of mercury, deuterium, and sodium are plotted with the measured angle of minimum deviation. Only the mercury and deuterium

Table 8: Sodium Wavelengths from a Prism

–	Red	Yellow	Green
Expected (nm)	615.43	589	568.27
Lab C (nm)	645	530.44	527.14
Percent Diff. (%)	15.45	10.46	12.65
Error (°)	0.083	0.083	0.083

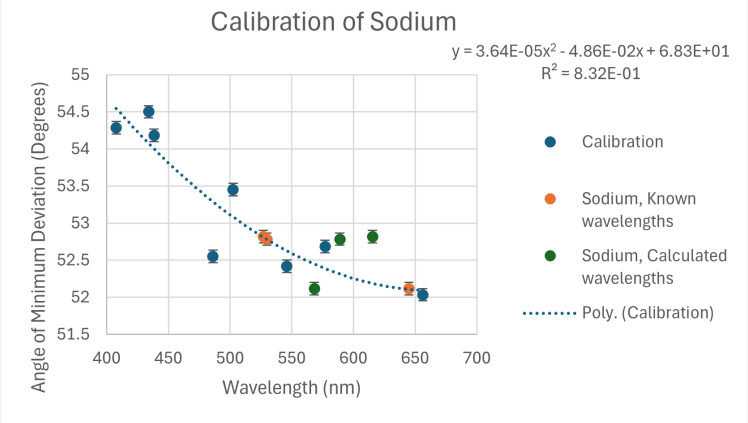


Figure 5: Calibration of sodium using mercury and deuterium

data points contribute to the trendline; the known sodium wavelengths are plotted for comparison purposes. The quadratic equation derived from the polynomial trendline in Figure 5 provides the calculated values of wavelengths λ. The calculated wavelength data points result from solving Equation 6 for wavelength λ using each measured angle of minimum deviation for sodium. A red (615.43 nm), yellow (one of the doublet, 589 nm) and green (568.27 nm) line was observed and calculated. The angles of these lines aided in finding the index of refraction. Table 9 features the index of refraction value n for each wavelength. The value for the index of refraction slightly varies based on the wavelength, since each wavelength travels at a different speed in a medium. It can be determined that the medium is of a glass-like composition, since glass can have an index of refraction of approximately 1.52-1.66 [13].

Figure 6 plots the data points, which are the index of refraction values, from Table 9. The graph has a relationship of the index of refraction n values versus wavelength. Two data sets are graphed: one data set are the known wavelengths of sodium with the calculated n values

Table 9: Index of Refraction

Color	Index of Refraction	Error
Red	1.67	0.083
Yellow	1.67	0.083
Green	1.66	0.083

and the second data set is the measured wavelengths with the calculated n values. Both data sets have a linear trendline, considering the n of the medium should be relatively consistent but not entirely flat. Although various wavelengths behave differently in the same medium, the composition of the medium remains constant. The error bars are 0.08333 degrees.

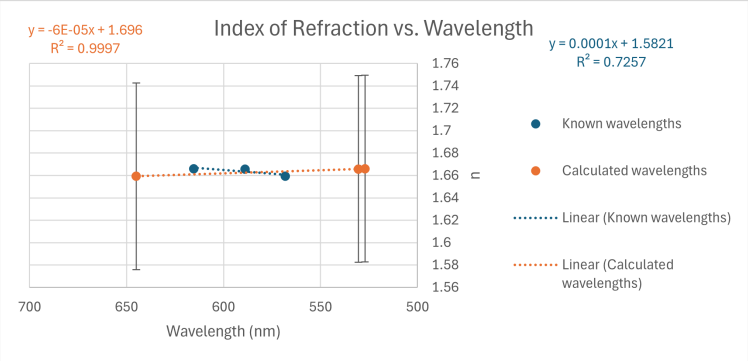


Figure 6: Index of Refraction

The values for the Rydberg constant of hydrogen are evident in Table 10. A Rydberg constant was produced from Lab B, Lab C, and Lab D data. Error is measured in the form of standard deviation, which is from the average of all Rydberg constants produced from the experimental wavelength values of a data set, and the percent difference of the expected and experimental values. Lab B yielded a percent difference of 1.65, Lab C has a difference of 22.85 percent, and Lab D has a difference of 13.22 percent. These constants are derived from the diffraction grating data.

Table 10: Rydberg Constant, Hydrogen

–	Expected (cm^{-1})	Calculated (cm^{-1})	STD (cm^{-1})	Percent Difference (%)
Lab B	109737	107912.014	5374.38	1.65
Lab C	109737	174747.97	16968.71	22.85
Lab D	109737	143185.71	22875.62	13.22

The values for the Rydberg constant of deuterium are displayed in Table 11. Due to being an isotope of hydrogen, deuterium has a balmer series and therefore, can produce a Rydberg constant. Error is measured in a similar manner to Table 11 with a standard deviation and percent difference Lab B yielded a percent difference of 1.21, Lab C has a difference of 41.65 percent, and Lab D has a difference of 34.26 percent. These constants are derived from the diffraction grating data.

Table 11: Rydberg Constant, Deuterium

–	Expected (cm^{-1})	Calculated (cm^{-1})	STD (cm^{-1})	Percent Difference (%)
Lab B	109737	111068.03	2130.7	1.21
Lab C	109737	167456.4	41502.12	41.65
Lab D	109737	155108	31839.64	34.26

Although the data featured for the prism results are from Lab C, Lab B and Lab D collected data with the prism as well. However, the prism data for Lab B and Lab C did not follow the procedure and does not suffice as accurate and relevant data for the objective of this experiment. This experiment aims to present data that is comparable to what is known; furthermore, the following data does not demonstrate the level of quality that is expected, since it was not collected or analyzed properly.

Table 12 displays the index of refraction calculated for sodium.

Table 12: Lab B Prism, Sodium

–	Index of refraction n	Minimum error for n	Maximum error for n
Yellow (Short)	1.65	0.0043	0.0046
Yellow (Long)	1.65	0.0054	0.0023
Red	1.64	0.0036	0.0053
Green	1.65	0.0049	0.005

Table 13 compiles the index of refraction values calculated for hydrogen.

The index of refraction values for the mercury gas-discharge lamp are seen in Table 14.

The error for n in the Lab B prism data was measured by calculating the index of refraction for a range of angles of a wavelength. A minimum and maximum angle is inserted into Equation 4 to produce a minimum and maximum n for each wavelength. The error is the

Table 13: Lab B Prism, Hydrogen

–	Index of refraction n	Minimum error for n	Maximum error for n
Red	1.64	0.004	0.004
Teal	1.66	0.0045	0.0031
Violet (Long)	1.67	0.0035	0.004
Violet (Short)	1.68	0.0039	0.0013

Table 14: Lab B Prism, Mercury

–	Index of refraction n	Minimum error for n	Maximum error for n
Yellow (Long)	1.65	0.005	0.005
Yellow (Short)	1.65	0.0039	0.0052
Green	1.65	0.0049	0.0044
Teal	1.65	0.015	0.0051
Blue	1.65	0.0049	0.004
Violet	1.67	0.0039	0.0047

difference between minimum and maximum n values and the index of refraction calculated for the observed angle. This data does not follow the procedure due to the incorrect approach to collecting the values of δ_{min}. Additionally, the incorrect method of analysis was executed. The index of refraction was calculated for each gas-discharge lamp data set instead of using two of the data sets for the calibration of the third set.

Similarly to Lab B, Lab C did not follow the procedure and measure the correct values of δ_{min}. Lab C produced a graph of the wavelength as a function of the angle, as seen in Figure 7.

A polynomial trendline is generated from the relationship between the known wavelengths and the angle of refraction. The analysis is not entirely correct because only the known wavelengths are displayed. No experimental measurements of wavelength were calculated using the quadratic equation from Figure 7. Another misstep is not calibrating one data set from the data of the two other sets. The index of refraction values could not be found from the incorrect data collection and method of analysis. The data and analysis from Lab D is preferred over that of Lab B and Lab C since it adheres to the expectations of the protocol. As a result, the data from Lab D is likely to be of better precision and accuracy. Regardless, the inclusion of the Lab B and Lab C data is necessary for emphasizing the errors that were

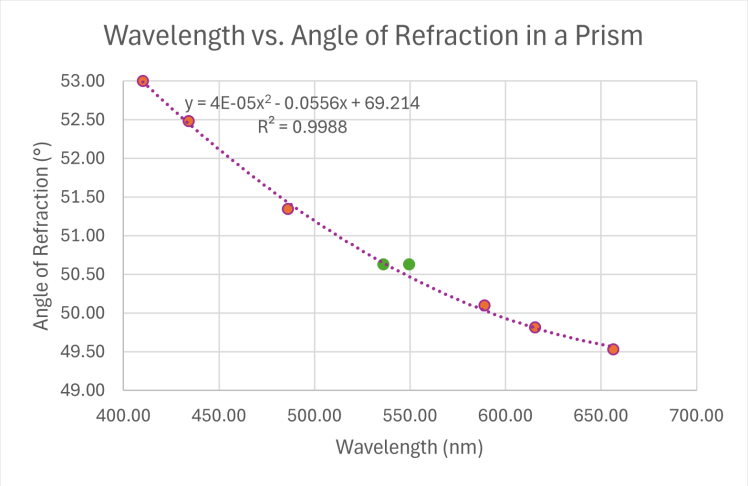


Figure 7: Index of Refraction

encountered in this experiment. The errors were influenced from lack of communication, disorganization, misjudgment of directions, and poor management of time.

5 Conclusion

The objective of this lab was to observe and measure the wavelengths of various gas-discharge lamps. The wavelengths of these gas-charge lamps were observed through a diffraction grating and a prism. Another goal was to derive the Rydberg constant from the grating data and the index of refraction from the prism data. Lab B, Lab C, and Lab D recorded data for different lamps but, aimed to fulfill the same objective. The prism data for Lab D was preferred over that of Lab B and Lab C due to the data collection process adhering to the correct method of finding the angles of minimum deviation, or the δ_{min} values. The most prevalent errors throughout the lab were a result of poor time management and lack of communication between lab partners. Poor time management impacted decision-making in the experimental process, which led to the misjudgement of directions from the protocol, disorganization, etc. Factors such as these may have influenced the obstacles encountered with collecting and analyzing the data, particularly with the prism.

Overall, it can be determined that the gas-discharge lamps used in this experiment

produced results that are comparable to what is characteristic and expected of the observed spectrum. The Rydberg constant for hydrogen was $107912.014 \pm 5374.38\ cm^{-1}$ for Lab B, $174747.97 \pm 16968.71\ cm^{-1}$ for Lab C, and $143185.71 \pm 22875.62\ cm^{-1}$ for Lab D. The percent difference ranged from 1.65-13.22 %. The Rydberg constant for deuterium was $111068.03 \pm 2130.7\ cm^{-1}$ for Lab B, $167456.4 \pm 41502.12\ cm^{-1}$ for Lab C, and $155108 \pm 31839.64\ cm^{-1}$ for Lab D. The percent difference ranged from 1.21-41.26 %. The index of refraction was calculated for sodium using the data from Lab C. The range of values for the index of refraction n is 1.66-1.67; this is approximately the determined n value for the medium of glass, which is 1.52-1.66 [13].

6 Distribution of Work

6.1 Lab B

- Credit to Aubrie Thoms
- Provided data and analysis for Lab B. The data sets that were relevant to the results of this lab were from a sodium, hydrogen, and deuterium gas-discharge lamp. Contributed a data set of sodium, hydrogen, and deuterium using an unknown grating and a set of sodium data from a known grating.
- Index of refraction values for sodium, hydrogen, and mercury. Method of collection and analysis were incorrect.
- Produced a Rydberg constant for Labs B, C, and D.
- Analyzed and contrived a graph from the Lab D prism data.
- Calibrated the Lab D data.
- Any alterations/changes made to the Lab C and Lab D files that were not made by the owners have been marked.

6.2 Lab C

- Credit to Jon Scholl
- Provided data and analysis for Lab C. The data sets that were relevant to the results of this lab were from a sodium, mercury, hydrogen, and deuterium gas-discharge lamp. Contributed a data set of sodium, hydrogen, and deuterium using an unknown grating and a set of sodium, hydrogen, and mercury data from a prism.

6.3 Lab D

- Credit to Jake Raugh
- Provided data and analysis for Lab D. The data sets that were relevant to the results of this lab were from a sodium, mercury, hydrogen, and deuterium gas-discharge lamp. Contributed a data set of mercury, hydrogen, and deuterium using an unknown grating and a set of sodium, deuterium, and mercury data from a prism.

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